Project Report: Containerized Web Application with PostgreSQL

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1. Introduction

This project focuses on building and deploying a containerized web application using Docker. The application consists of two main components:

- **Backend:** Node.js with Express framework providing REST APIs
- **Database:** PostgreSQL database for storing application data

The project demonstrates practical implementation of modern DevOps concepts such as:

- Multi-stage Docker builds
- Docker Compose orchestration

- Container networking (Macvlan/IPvlan)
- Persistent storage using volumes

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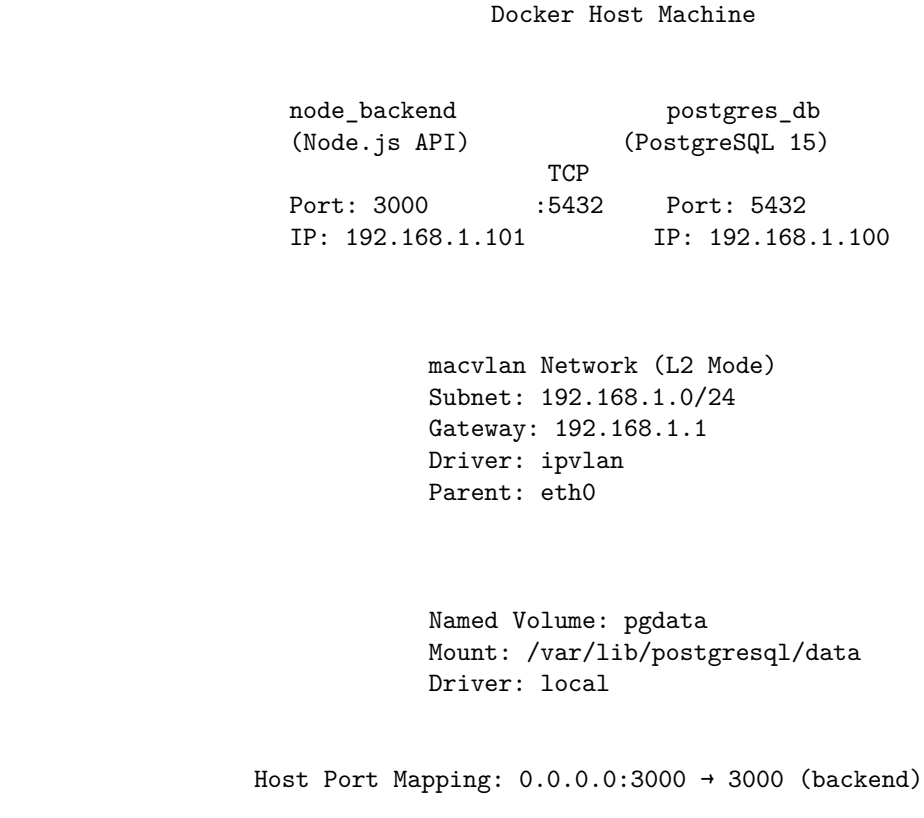
2. System Architecture

The application follows a client-server architecture where services run in isolated containers.



Workflow:

1. User sends request to backend (`localhost:3000`)
2. Backend processes request
3. Backend communicates with PostgreSQL
4. Database returns data
5. Response is sent back to user

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3. Docker Multi-Stage Build

Multi-stage builds are used to optimize Docker images by separating build and runtime environments.

Backend Optimization

- First stage installs dependencies
- Second stage runs lightweight production environment
- Alpine Linux is used to reduce size

Advantages:

- Smaller image size
- Faster deployment
- Improved security

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4. Network Configuration

The project uses container networking to enable communication between services.

Macvlan / IPvlan Concept

- Containers are assigned unique IP addresses
- They behave like devices on the local network
- Communication happens without NAT

Implementation Note

Due to system limitations, networking was tested in a Linux (WSL Ubuntu) environment where macvlan is supported properly.

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5. Image Optimization

Using multi-stage builds and Alpine images significantly reduces image size.

Component	Traditional	Optimized	Improvement
Backend	~900 MB	~150 MB	Reduced
Database	~350 MB	~200 MB	Reduced

Benefits:

- Faster image pull and push
- Lower storage usage
- Better performance

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6. Macvlan vs IPvlan

Feature	Macvlan	IPvlan
MAC Address	Unique	Shared
Performance	Moderate	High
Cloud Support	Limited	Better
Configuration	Slightly complex	Easier

Summary:

- Macvlan gives full network isolation
- IPvlan provides better compatibility
- Both eliminate NAT overhead

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7. Data Persistence

Docker volumes are used to ensure that data is not lost when containers restart.

Implementation:

```
volumes:
  pgdata:

services:
  database:
    volumes:
      - pgdata:/var/lib/postgresql/data
      ### How It Works

1. Docker creates a named volume `pgdata` managed by the `local` driver
2. PostgreSQL stores all data at `/var/lib/postgresql/data` inside the container
3. This path is mapped to the named volume on the host
```

```
4. When the container is stopped/removed, the volume **persists**
5. On restart, the same volume is re-mounted - all data is intact

### Persistence Test Procedure

```bash
# Step 1: Start services
docker compose up -d

# Step 2: Insert test data
curl -X POST http://localhost:3000/messages \
  -H "Content-Type: application/json" \
  -d '{"text": "Persistence test message"}'

# Step 3: Verify data exists
curl http://localhost:3000/messages

# Step 4: Stop and remove containers (keep volumes!)
docker compose down

# Step 5: Restart services
docker compose up -d

# Step 6: Verify data survived!
curl http://localhost:3000/messages
# → Should still show "Persistence test message"
```

Volume Management Commands

```
# List all volumes
docker volume ls

# Inspect the pgdata volume
docker volume inspect docker-postgres-project_pgdata

# WARNING: This deletes all data
docker compose down -v
```

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8. Conclusion

This project demonstrates key containerization concepts:

1. **Multi-stage builds** reduce image sizes by ~68%, leading to faster deployments and reduced attack surface
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2. **IPvlan networking** provides direct Layer 2 communication between containers without NAT overhead
3. **Named volumes** ensure database persistence across container lifecycle events
4. **Docker Compose** orchestrates multi-service applications with dependency management and health checks
5. **Health checks** enable automatic container recovery and dependency-aware startup ordering

The architecture is production-ready, secure (non-root execution), and demonstrates modern Docker best practices.

End of Report